COS 201

COMPUTER PROGRAMMING I

WEEK 1, WEEK 2 & WEEK 3 ASSIGNMENTS

Student Name	Omage Ikhide Divine
Course	COS 201 — Computer Programming I
Programme	Software Engineering
Matriculation Number	To be filled in by student 25/18644

Prepared from the COS 201 lecture note for Weeks 1–3. The document includes written answers, Java source code, sample outputs, README text, and submission-message templates.

Important: The GitHub repository must be created and the files must be uploaded by the student. The repository URL and matriculation number cannot be filled in accurately until those details exist.

CONTENTS

1. Week 1 — First Java program and written problem-solving activities

2. Week 2 — Development environment and first Java program

3. Week 3 — Variables, data types, expressions, input/output and practical programs

4. README text and submission checklists

The code is formatted for copying into files with the exact filenames shown. Java is case-sensitive; ensure the filename matches the public class name.

WEEK 1 — INTRODUCTION TO PROGRAMMING AND PROBLEM-SOLVING

A. Practical: FirstProgram.java

Required folder: **COS201-Programming-I/week01/FirstProgram.java**. This version displays the course, programme, three reasons programming matters to Software Engineering, and one personal learning goal.

```
public class FirstProgram {
    public static void main(String[] args) {
        System.out.println("Course: COS 201 - Computer Programming I");
        System.out.println("Programme: Software Engineering");
        System.out.println("Reason 1: Programming helps solve computational problems.");
        System.out.println("Reason 2: Programming supports automation of tasks.");
        System.out.println("Reason 3: Programming enables software and applications to be developed.");
        System.out.println("My goal: I want to write, test, and understand useful Java programs.");
    }
}
```

Expected output

```
Course: COS 201 - Computer Programming I
Programme: Software Engineering
Reason 1: Programming helps solve computational problems.
Reason 2: Programming supports automation of tasks.
Reason 3: Programming enables software and applications to be developed.
My goal: I want to write, test, and understand useful Java programs.
```

Compile and run from the project folder:

```
cd week01
javac FirstProgram.java
java FirstProgram
```

B. Activity A — Identify Input, Process and Output (IPO)

1. Area of rectangle (length 12, width 5)

Input: length = 12, width = 5.

Process: area = length × width = 12 × 5.

Output: Area = 60 square units.

2. Average of four examination scores

Input: four scores, score1–score4.

Process: total = score1 + score2 + score3 + score4; average = total ÷ 4.

Output: The average score.

3. Classify a number

Input: one number n.

Process: if n > 0, classify as positive; if n < 0, negative; otherwise, zero.

Output: Positive, negative, or zero.

4. Total price of six items at ■250 each

Input: quantity = 6; unit price = ■250.

Process: total = quantity × unit price = 6 × ■250.

Output: ■1,500.

5. Count failed login attempts in a sample log

Input: a sample security log containing login results.

Process: examine each recorded attempt and increase the count whenever the result is a failed login.

Output: Total number of failed login attempts.

WEEK 1 — WRITTEN ACTIVITIES (CONTINUED)

C. Activity B — Algorithms

Accept two numbers and display their sum

1. Start.

2. Input first number A and second number B.

3. Calculate SUM = A + B.

4. Display SUM.

5. Stop.

Calculate the area of a circle

1. Start.

2. Input radius r.

3. Calculate AREA = π × r × r.

4. Display AREA.

5. Stop.

Check whether a student passed (pass mark 50)

1. Start.

2. Input student score.

3. If score ≥ 50, display “Pass”; otherwise display “Fail”.

4. Stop.

Display the largest of three numbers

1. Start.

2. Input A, B and C.

3. If A ≥ B and A ≥ C, largest = A; else if B ≥ A and B ≥ C, largest = B; otherwise largest = C.

4. Display largest.

5. Stop.

Calculate total cost of purchased items

1. Start.

2. Input quantity and unit price.

3. Calculate TOTAL = quantity × unit price.

4. Display TOTAL.

5. Stop.

D. Activity C — Pseudocode and flowcharts

Selected problem 1: Add two numbers.

```
BEGIN
    INPUT A
    INPUT B
    SUM ← A + B
```

```
    OUTPUT SUM
END
```

Text flowchart (use oval for Start/Stop, parallelograms for Input/Output, and a rectangle for processing):

```
  ( START )
      |
/ Input A, B /
      |
[ SUM = A + B ]
      |
 / Output SUM /
      |
   ( STOP )
```

Selected problem 2: Pass or fail using 50 as the pass mark.

```
BEGIN
    INPUT score
    IF score >= 50 THEN
        OUTPUT "Pass"
    ELSE
        OUTPUT "Fail"
    ENDIF
END
```

Text flowchart (use a diamond for the decision):

```
        ( START )
            |
      / Input score /
            |
      < score >= 50? >
        /         \
     Yes           No
      |             |
/ Output Pass / / Output Fail /
        \         /
           |
        ( STOP )
```

E. Week 1 GitHub checklist

- Create or use the repository named COS201-Programming-I.
- Save the source file as week01/FirstProgram.java inside the cloned repository.
- Update README.md with the student details and a description of the Week 1 practical.
- Compile and run the program; check that the output matches the expected output.
- Commit the changes with the summary: Complete Week 1 Java practical.
- Push the commit to GitHub and verify that the files appear online.
- Submit the actual repository URL through the lecturer's submission form.

WEEK 2 — DEVELOPMENT ENVIRONMENT AND FIRST JAVA PROGRAM

A. HelloWorld.java

Required path: **week02/HelloWorld.java**. The lecture note's basic program is personalized with the student's course details.

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("Welcome to COS 201.");
        System.out.println("I am learning Java programming.");
        System.out.println("Student: Omage Ikhide Divine");
        System.out.println("Programme: Software Engineering");
    }
}
```

Expected output

```
Hello, World!
Welcome to COS 201.
I am learning Java programming.
Student: Omage Ikhide Divine
Programme: Software Engineering
```

Compile and run inside the week02 directory:

```
javac HelloWorld.java
java HelloWorld
```

B. StudentProfile.java — Additional practice file

The Week 3 repository layout in the lecture note lists StudentProfile.java under week02. This simple file collects and displays the student's information.

```
import java.util.Scanner;

public class StudentProfile {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter student name: ");
        String name = input.nextLine();

        System.out.print("Enter programme: ");
        String programme = input.nextLine();

        System.out.print("Enter level: ");
        int level = input.nextInt();

        System.out.println("\n--- STUDENT PROFILE ---");
        System.out.println("Name: " + name);
        System.out.println("Programme: " + programme);
        System.out.println("Level: " + level);

        input.close();
    }
}
```

C. Short explanations

• **JDK:** Java Development Kit; includes the compiler and tools needed to develop Java programs.

• **Compiler (javac):** Converts Java source code into bytecode.

• **JVM:** Java Virtual Machine; executes Java bytecode.

• **Git:** Tracks changes to files over time.

• **GitHub:** Hosts repositories online.

• **GitHub Desktop:** Provides a graphical interface for committing and pushing changes.

• **Commit:** Saves a snapshot of changes in the local repository.

• **Push:** Sends committed changes to the online repository.

D. README.md — Week 2 section

```
# COS 201 – Computer Programming I

## Student Information
- Name: Omage Ikhide Divine
- Matriculation Number: [Enter your matric number]
- Programme: Software Engineering
- Course: COS 201
- Week: 2

## Practical Title
Development Environment and First Java Program

## Objective
To configure the Java development environment, write a simple Java program,
compile it, execute it and submit the source code through GitHub.

## Tools Used
- Java Development Kit (JDK)
- Visual Studio Code
- Git
- GitHub Desktop
- GitHub

## Files
- week02/HelloWorld.java
- week02/StudentProfile.java

## How to Run
In the week02 directory:
1. javac HelloWorld.java
2. java HelloWorld
3. javac StudentProfile.java
4. java StudentProfile

## Learning Outcome
I practised creating, compiling and running Java programs and preparing
source code for submission through GitHub.
```

E. Week 2 submission message

```
COS 201 – Week 2 Practical Submission

Dear Lecturer,

I am submitting my Week 2 practical assignment for COS 201 – Computer
Programming I.

Student Name: Omage Ikhide Divine
Matriculation Number: [Enter your matric number]
Programme: Software Engineering
Assignment: Development Environment and First Java Program
GitHub Repository: [Paste your actual repository URL]

I have included my Java source code and README documentation in the
repository. I have also compiled and executed the program to verify the
output.

Thank you.
```

Before submission, replace the matric-number and repository-link placeholders with your actual details. Only say the programs were compiled and tested after you have run them successfully.

WEEK 3 — VARIABLES, DATA TYPES, EXPRESSIONS AND INPUT/OUTPUT

A. SimpleCalculator.java

Calculates the sum, difference, product and quotient of two numbers, checking for division by zero.

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the first number: ");
        double firstNumber = input.nextDouble();

        System.out.print("Enter the second number: ");
        double secondNumber = input.nextDouble();

        double sum = firstNumber + secondNumber;
        double difference = firstNumber - secondNumber;
        double product = firstNumber * secondNumber;

        System.out.printf("Sum: %.2f%n", sum);
        System.out.printf("Difference: %.2f%n", difference);
        System.out.printf("Product: %.2f%n", product);

        if (secondNumber != 0) {
            double quotient = firstNumber / secondNumber;
            System.out.printf("Quotient: %.2f%n", quotient);
        } else {
            System.out.println("Division is undefined for a zero divisor.");
        }

        input.close();
    }
}
```

Sample run

```
Enter the first number: 20
Enter the second number: 4
Sum: 24.00
Difference: 16.00
Product: 80.00
Quotient: 5.00
```

B. AverageCalculator.java

Reads a student's name and three scores, then calculates the total and average.

```
import java.util.Scanner;

public class AverageCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter student name: ");
        String name = input.nextLine();

        System.out.print("Enter score for Course 1: ");
        double score1 = input.nextDouble();

        System.out.print("Enter score for Course 2: ");
        double score2 = input.nextDouble();

        System.out.print("Enter score for Course 3: ");
        double score3 = input.nextDouble();

        double total = score1 + score2 + score3;
        double average = total / 3.0;

        System.out.println("\n--- STUDENT RESULT ---");
```

```
        System.out.println("Student: " + name);
        System.out.printf("Total: %.2f%n", total);
        System.out.printf("Average: %.2f%n", average);

        input.close();
    }
}
```

Sample run

```
Enter student name: Omage Ikhide Divine
Enter score for Course 1: 70
Enter score for Course 2: 80
Enter score for Course 3: 90

--- STUDENT RESULT ---
Student: Omage Ikhide Divine
Total: 240.00
Average: 80.00
```

WEEK 3 — PRACTICAL PROGRAMS (CONTINUED)

C. ItemCost.java

Calculates the total cost of items using quantity × unit price.

```
import java.util.Scanner;

public class ItemCost {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter item name: ");
        String itemName = input.nextLine();

        System.out.print("Enter quantity: ");
        int quantity = input.nextInt();

        System.out.print("Enter unit price: ");
        double unitPrice = input.nextDouble();

        double totalCost = quantity * unitPrice;

        System.out.println("\n--- PURCHASE SUMMARY ---");
        System.out.println("Item: " + itemName);
        System.out.println("Quantity: " + quantity);
        System.out.printf("Unit price: %.2f%n", unitPrice);
        System.out.printf("Total cost: %.2f%n", totalCost);

        input.close();
    }
}
```

Sample run

```
Enter item name: Notebook
Enter quantity: 3
Enter unit price: 1500
--- PURCHASE SUMMARY ---
Item: Notebook
Quantity: 3
Unit price: 1500.00
Total cost: 4500.00
```

D. How to compile and run

```
# Run each command from inside the week03 directory
javac SimpleCalculator.java
java SimpleCalculator

javac AverageCalculator.java
java AverageCalculator

javac ItemCost.java
java ItemCost
```

E. Concepts used

• **Variable:** A named storage location for a value, e.g. double unitPrice = 1500.0;

• **Data type:** Specifies the kind of value a variable can hold, e.g. int, double, String, boolean.

• **Expression:** A combination of values and operators that produces a result, e.g. quantity * unitPrice.

• **Scanner:** A Java class used to read keyboard input.

• **Arithmetic operators:** + addition, - subtraction, * multiplication, / division, % remainder.

• **Relational operators:** Compare values, e.g. >=, <=, ==, !=.

• **Logical operators:** && means AND; || means OR; ! means NOT.

• **Formatted output:** %.2f displays a decimal number to two places; %n creates a new line.

F. Testing checklist

• SimpleCalculator: test ordinary numbers and also enter 0 as the second number to check the zero-divisor message.

• AverageCalculator: test three different scores and confirm total ÷ 3 gives the average.

• ItemCost: test a quantity and unit price where the multiplication can be checked manually.

• Check that each file compiles without errors and that output matches the expected calculations.

WEEK 3 — README AND SUBMISSION

A. Add this section to README.md

```
## Week 3: Variables, Data Types, Expressions and Input/Output

### Student Details
- Name: Omage Ikhide Divine
- Matriculation Number: [Enter your matric number]
- Programme: Software Engineering
- Course: COS 201 — Computer Programming I

### Objectives
- Declare and initialize Java variables.
- Use appropriate primitive and reference data types.
- Perform arithmetic calculations.
- Read keyboard input using Scanner.
- Display formatted output.
- Compile, test and submit Java programs using GitHub.

### Programs
1. week03/SimpleCalculator.java
   - Performs arithmetic calculations on two numbers.
2. week03/AverageCalculator.java
   - Calculates the total and average of three scores.
3. week03/ItemCost.java
   - Calculates total cost using quantity and unit price.

### How to Run
Open a terminal in the week03 directory.
- javac SimpleCalculator.java
- java SimpleCalculator
- javac AverageCalculator.java
- java AverageCalculator
- javac ItemCost.java
- java ItemCost

### Testing
Test each program with different valid inputs and verify that the results
match the expected calculations.
```

B. Week 3 submission message

```
COS 201 — Week 3 Practical Submission

Dear Lecturer,

I am submitting my Week 3 practical work on Variables, Data Types,
Expressions and Input/Output.

Student Name: Omage Ikhide Divine
Matriculation Number: [Enter your matric number]
Programme: Software Engineering
Course: COS 201 — Computer Programming I
Week: 3
GitHub Repository URL: [Paste your actual repository URL]

Programs submitted:
- SimpleCalculator.java
- AverageCalculator.java
- ItemCost.java

I have saved the source code, tested the programs and pushed the required
files to GitHub. The README file contains instructions for compiling and
running the programs.

Thank you.
```

C. Final project structure

```
COS201-Programming-I/
■■■ README.md
■■■ week01/
■   ■■■ FirstProgram.java
■■■ week02/
■   ■■■ HelloWorld.java
■   ■■■ StudentProfile.java
■■■ week03/
    ■■■ SimpleCalculator.java
    ■■■ AverageCalculator.java
    ■■■ ItemCost.java
```

This is the intended structure. Keep only files that you have actually created and tested. The lecturer's notes require the actual repository URL to be submitted; this PDF is not a substitute for pushing the source files to GitHub.

D. Final checklist

- Fill in your matriculation number wherever required.
- Create the files using the exact filenames and folders in this PDF.
- Run each Java program in the terminal and fix any compilation errors.
- Update the main README.md so it accurately describes the files in your repository.
- Commit your changes in GitHub Desktop, then push them to GitHub.
- Open the online repository and confirm that all expected files are visible.
- Copy your real repository URL and submit it using the lecturer's form.